

Bug Fix Impact Analysis

METIMMOX-1 Economic Evaluation

Ben Geisler

2025-11-30

Table of contents

1	Summary	1
2	Metadata Comparison	2
2.1	Global Parameters	2
3	Base Case Results Comparison	3
3.1	Efficiency Frontier Comparison	3
3.2	Net Monetary Benefit Comparison	3
4	PSA Results Comparison	4
4.1	PSA Visual Analysis	5
4.1.1	PSA Scatter Plots (Side-by-Side)	5
4.1.2	PSA Scatter Plots (Overlaid)	6
5	Comparison Across Recent Issues	7
5.1	Available Snapshot Pairs	7
5.2	Multi-Issue Impact Summary	8
6	Summary	9

1 Summary

This report compares cost-effectiveness analysis results before and after addressing GitHub issue #76. The comparison assesses whether bug fixes or model changes affected results.

Current Comparison:

- **Issue:** #76 - [P0] PFS Can Exceed OS (Biological Impossibility Not Enforced)
- **Baseline:** Commit 64cf543 (2025-11-30 13:33:10)
- **Fixed:** Commit 64cf543 (2025-11-30 14:47:30)

2 Metadata Comparison

Table 1: Metadata Comparison

Item	Baseline	Fixed
Git Commit	64cf543	64cf543
Timestamp	2025-11-30 13:33:10	2025-11-30 14:47:30
R Version	R version 4.3.2 (2023-10-31 ucrt)	R version 4.3.2 (2023-10-31 ucrt)
Issue Number	76	76

2.1 Global Parameters

Table 2: Global Parameter Comparison

	Parameter	Baseline	Fixed	Changed
WTP	WTP	51,000	51,000	No
time_horizon	time_horizon	520	520	No
n_samples	n_samples	5,000	5,000	No
n_sim	n_sim	5,000	5,000	No
cl	cl	0.01923077	0.01923077	No
dr	dr	0.04	0.04	No
DSA_mult	DSA_mult	0.2	0.2	No
USE_BOTH_MODELS	USE_BOTH_MODELS	0	0	No

3 Base Case Results Comparison

Table 3: Base Case Incremental Cost-Effectiveness Comparison

Strategy	Incremental Cost		Incremental Effect		ICER	
	BL Δ Cost (€)	Fix Δ Cost (€)	BL Δ Effect (QALYs)	Fix Δ Effect (QALYs)	BL ICER (€/QALY)	Fix ICER (€/QALY)
control	0.00	0.00	0.0000	0.0000	—	—
crp	33,930.80	33,930.80	-0.0599	-0.0599	—	—
tlr	64,024.26	64,024.26	-0.1406	-0.1406	—	—
tmb_braf	41,540.41	41,540.41	0.0252	0.0252	1,645,887	—

3.1 Efficiency Frontier Comparison

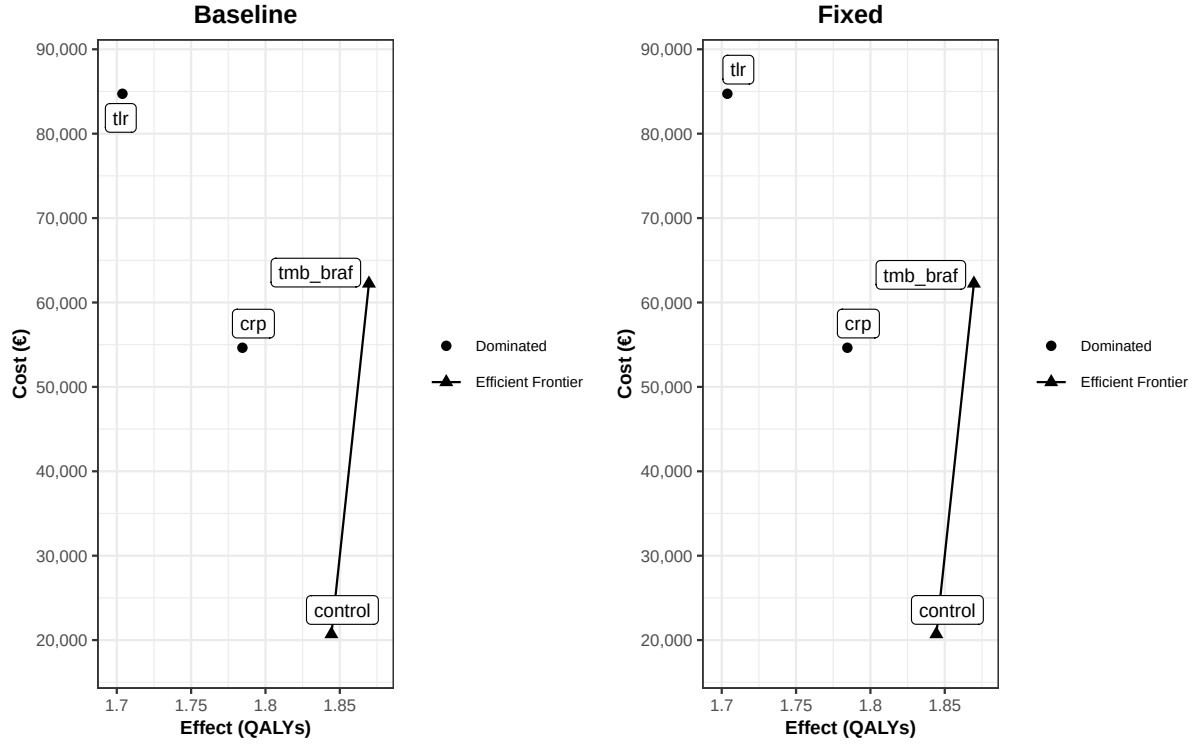


Figure 1: Cost-effectiveness efficiency frontier comparison using dampack. Baseline (left) and fixed (right) strategies are shown with the efficiency frontier connecting non-dominated strategies. Dominated strategies appear off the frontier line.

3.2 Net Monetary Benefit Comparison

Table 4: Net Monetary Benefit Comparison at WTP = €51,000

	Strategy	Baseline (€)	Fixed (€)	Difference (€)
1	control	73,356.74	73,356.74	0.00
2	crp	36,372.76	36,372.76	0.00
4	tmb_braf	33,103.52	33,103.52	0.00
3	tlr	2,163.83	2,163.83	0.00

Optimal Strategy: Unchanged (control)

4 PSA Results Comparison

Table 5: PSA Cost Results (n = 5,000 simulations)

Strategy	Baseline Mean (2.5%; 97.5%)	Fixed Mean (2.5%; 97.5%)	Difference (€)
control	€20,941.26 (€15,716.53; € 27,097.95)	€20,937.05 (€15,884.01; € 26,893.85)	-4.22
crp	€54,777.87 (€41,404.79; € 70,197.13)	€54,688.84 (€41,335.55; € 70,131.38)	-89.03
tlr	€87,148.64 (€62,222.38; €115,922.50)	€86,970.26 (€62,500.02; €115,792.02)	-178.38
tmb_braf	€65,689.73 (€48,558.82; € 85,701.21)	€65,594.08 (€48,472.98; € 85,668.50)	-95.64

Table 6: PSA Effects Results (n = 5,000 simulations)

Strategy	Baseline Mean (2.5%; 97.5%)	Fixed Mean (2.5%; 97.5%)	Difference (QALYs)
control	1.7029 (1.1074; 2.3870)	1.6975 (1.0970; 2.3560)	-0.0054
crp	1.8788 (1.2574; 2.4069)	1.8741 (1.2365; 2.4252)	-0.0047
tlr	1.8091 (1.1858; 2.3680)	1.8038 (1.1685; 2.3605)	-0.0052
tmb_braf	1.9828 (1.2947; 2.6785)	1.9756 (1.2567; 2.6360)	-0.0071

Table 7: PSA ICER Results (n = 5,000 simulations)

Strategy	Baseline Mean ICER	Fixed Mean ICER
control	—	—
crp	€192,385/QALY	€191,159/QALY
tlr	€623,716/QALY	€621,117/QALY
tmb_braf	€159,890/QALY	€160,569/QALY

Note: ICERs calculated from PSA mean costs and effects as point estimates.

4.1 PSA Visual Analysis

4.1.1 PSA Scatter Plots (Side-by-Side)

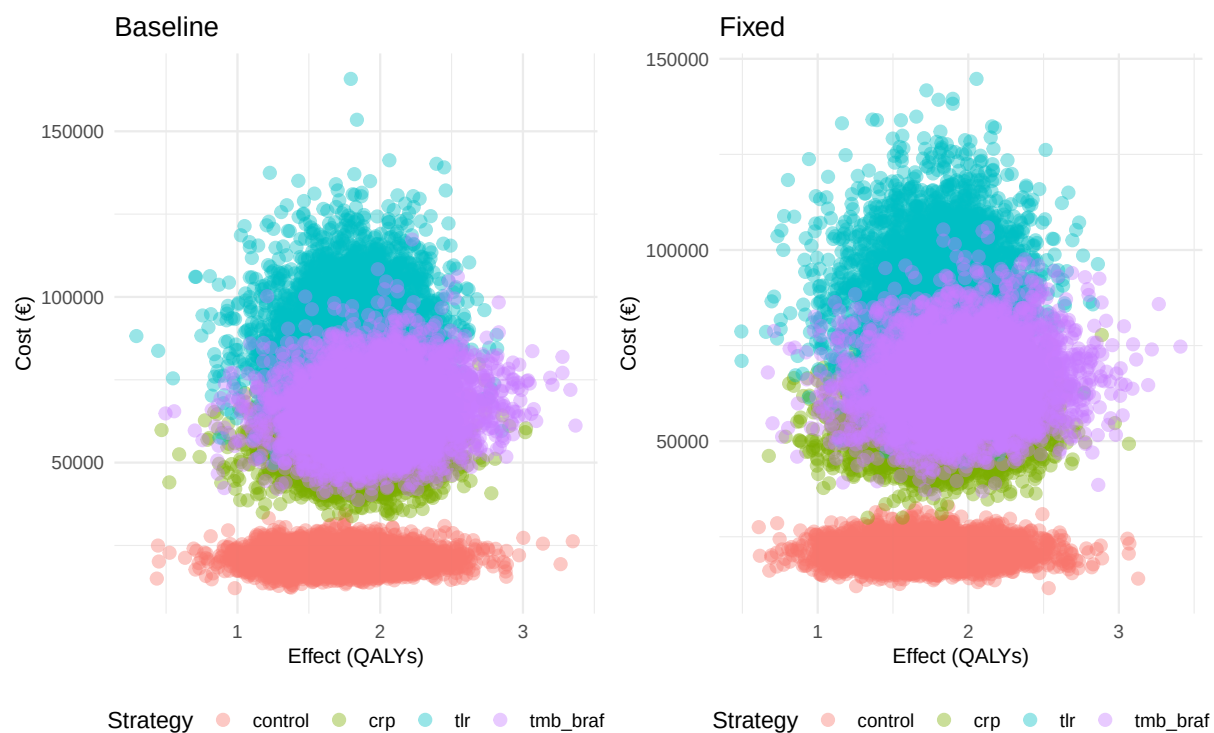


Figure 2: Cost-effectiveness plane showing PSA simulations baseline (left) and fixed (right)

4.1.2 PSA Scatter Plots (Overlaid)

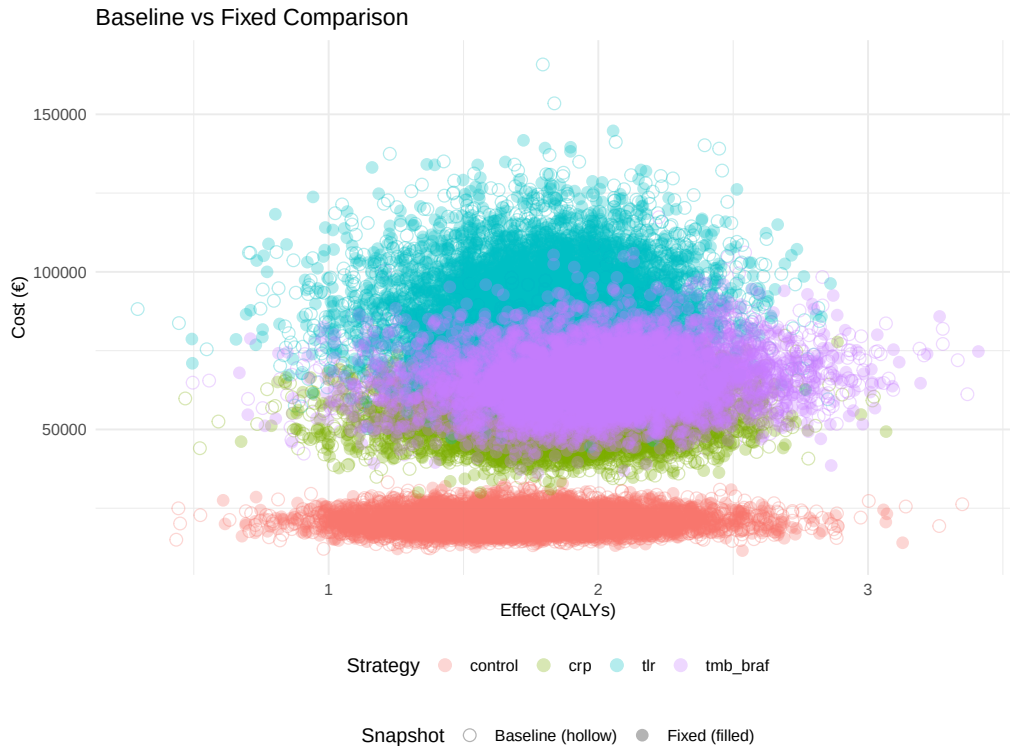


Figure 3: Overlaid cost-effectiveness plane comparing baseline and fixed PSA simulations. This plot overlays all PSA simulation results from both snapshots on a single plane, where each point represents one PSA iteration for a given strategy. Hollow circles indicate baseline simulations, while filled circles indicate fixed simulations. Colors distinguish strategies, while shapes distinguish snapshots. The overlay allows direct visual comparison of how the probabilistic distributions changed after the bug fix: if the fix was purely methodological with no impact on central estimates, baseline and fixed points would overlap completely, whereas systematic shifts in the point clouds indicate changes in underlying model behavior.

5 Comparison Across Recent Issues

5.1 Available Snapshot Pairs

Table 8: All Available Snapshot Pairs

	Issue	Title	Baseline Commit	Fixed Commit	Status
7	#76	[P0] PFS Can Exceed OS (Biological Impossibility Not Enforced)	64cf543	64cf543	Current
9	#87	[P0] Fix PSA methodology: Replace single-patient sampling with subgroup population averaging	42d4acb	0a15902	
8	#86	[P1] Mltpl documentation issues	ee96c31	f3e5604	
6	#70	PSA uses inconsistent methods for control vs. biomarker strategy predictions	998dacd	605ff90	
5	#68	[P4] Include Weibull model for reference	4665f93	998dacd	
1	#42	[P0] Critical Bug: Incorrect death transition calculation in calculate_outcomes.R	607b539	cadf81a	
4	#67	[P4] Move snapshot folder	f3674e8	d5d1356	
2	#62	[P4] Move cached file	d046c7c	f3674e8	
3	#64	[P3] impact of bug fixes	8681d1e	8681d1e	

5.2 Multi-Issue Impact Summary

This table summarizes the impact of bug fixes across the most recent issues:

Table 9: Summary of Bug Fix Impacts (5 Most Recent Issues)

Issue	Title	Opt. (BL)	Opt. (Fix)	Chg?	Max Cost Δ (€)	Max Effect Δ (QALYs)	Max ICER Δ (€/QALY)
#76	[P0] PFS Can Exceed OS (Biological Impossibility Not Enforced)	control	control	No	0.00	0.0000	0
#87	[P0] Fix PSA methodology: Replace single-patient sampling with subgroup population averaging	control	control	No	0.00	0.0000	0
#86	[P1] Mltpl documentation issues	control	control	No	0.00	0.0000	0
#70	PSA uses inconsistent methods for control vs. biomarker strategy predictions	control	control	No	8,290.42	0.4081	-Inf
#68	[P4] Include Weibull model for reference	control	control	No	0.00	0.0000	0

6 Summary

The comparison for issue #76 shows:

- **Optimal Strategy:** Unchanged (control)
- **Maximum Cost Change:** €0.00 (minor)
- **Maximum Effect Change:** 0.000 QALYs (minor)
- **Maximum ICER Change:** €0/QALY (minor)
- **Parameter Changes:** None detected

Interpretation: Changes are within expected bounds for bug fixes that improve numerical stability or code quality.

Report completed on: 2025-11-30

Repository: ben-geisler/METIMMOX-1

Report version: 1.3